

Shashank S

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Experience

Senior Machine Learning Engineer

Walmart · Dec 2024 - Jul 2025 · Transitioned to full-time graduate study

- Optimized PySpark jobs in the adtech ML pipeline — profiled bottlenecks, rewrote shuffle-heavy transformations, and scaled data processing to keep pace with growing ad volume without proportional infrastructure cost
- Engineered a Text2SQL agentic system so non-technical teams could query data in plain English, with associated visualizations and active chat interface
- Built an ML observability framework for production model monitoring; set up GitHub Actions pipelines standardizing ML build, test, and deployment workflows

Machine Learning Engineer III

Glance/InMobi · May 2021 - Nov 2024 · Promoted: Data Scientist II → MLE III, Jul 2022

- Served personalized content to a **10M DAU cohort at sub-50ms p99 latency** via an end-to-end real-time ML pipeline — prediction service, Aerospike feature store, Kafka event streams, ELK logging, and Airflow-managed retraining
- Built and iterated recommendation models (multi-armed bandits, GBMs, graph-based collaborative filtering) — drove **5% engagement lift** and **12% conversion** from sparse to dense users
- Co-authored lookalike modeling research for cold-start users: SVD + K-Means on user metadata seeding a MAB model — **+11.68% time-spent overall, +18.38% for cold users**
- Owned the ML automation platform end-to-end: streaming pipelines, model health controllers, automated rollback triggers, and event-driven personalization — cut oncall response time significantly by removing humans from routine incident loops
- Drove A/B testing infrastructure and experimentation culture across the org

Data Scientist

Bright Money · Jun 2019 - Apr 2021 · Promoted: Junior Research Scientist → Data Scientist, Jan 2021

- Founding data scientist at an early-stage fintech — built the ML layer from scratch across the core product
- Shipped *Debt Manager*: end-to-end financial analysis and automated payment scheduling, **10K+ users, \$100K+ daily transaction volume**

- Built bill identification, affordability scoring, and overdraft prediction models — zero-to-one, no prior art internally
- Forecasting models for income, expenses, and account balances underpinned the product's core retention loop; model outputs surfaced directly in user-facing features

Education

Master of Science, Computer Science

Georgia Institute of Technology, Atlanta, USA · May 2026

- Specialization: *High Performance Computing* · GPA 3.87/4.0
- Research: Weight drift correction in Analog ML accelerators with *Prof. Hyesoon Kim* · paper in preparation (targeting MICRO 2026)
- *Relevant Coursework*: GPU Hardware & Software, Scientific Machine Learning, High Performance Computing, High Perf Parallel Computing, HPC Architecture, Deep Learning, Compiler Design, Designing for Curiosity

Bachelor of Engineering, Computer Science & Engineering

Sir M. Visvesvaraya Institute of Technology, Bengaluru, India · 2018

- First Class with Distinction — 79.8% · *Thesis: Image Regeneration with Generative Models*
- *Electives*: Pattern Recognition, Clouds & Clusters, Artificial Intelligence

Skills

Languages: Python, C/C++, PySpark, SQL, Go, Shell, JavaScript

Machine Learning & AI: PyTorch, scikit-learn, XGBoost, LightGBM, Recommendation Systems, LLMs, LangChain, Agentic Workflows, Deep Learning, NLP, MLOps, Feature Stores, A/B Testing

High Performance Computing: CUDA, MPI, OpenMP, GPU Profiling, Distributed Systems, Scientific ML (PINNs)

Infrastructure: Spark, Kafka, Docker, Kubernetes, Airflow, Argo, Aerospike, Prometheus, Grafana, Neptune

Cloud & Tools: AWS, GCP, Azure, PostgreSQL, Git, LaTeX, Streamlit, Linux

Research

Weight Drift Correction for Edge ML Accelerators

- *Paper in preparation, targeting MICRO 2026*
- Algorithmic approaches to correcting drift errors in analog MVM crossbar arrays; invertible matrix scaling, SVD-based time-invariant decomposition, and RL-based

active correction; evaluated end-to-end on ResNet18/ImageNet under SONOS and PCMDrift models

Automated Debt Management

- *USPTO (US20220261886A1, US20230075411A1)*
- R&D contributor to the systems underlying these patents

Deep Learning Segmentation and Quantification of Meibomian Glands

- *Biomedical Signal Processing and Control*
- CNN-based segmentation strategy for ophthalmic imaging

Image Regeneration with Generative Models

- Undergrad thesis
- Use CapsNet as a discriminator in Generative Adversarial Network and demonstrate its application using semantic inpainting on MNIST and face images.

Honors & Extras

ACM ICPC 2015 — Honorable Mention

Letter of Commendation, Minister of HRD — AISSCE Board Exam

Section Leader, Code in Place · Stanford University · 2021, 2023, 2024

Taught Python to 30+ students per cohort

Certifications Neural Networks & Deep Learning (deeplearning.ai) · Machine Learning (Stanford / Coursera) · Android Developer Nanodegree (Google / Udacity)